

UNIVERSITY MEDICAL CENTER

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CLIA #22D1234567 · CAP #12345678

SURGICAL PATHOLOGY REPORT

Patient Name: Jenny M
Date of Birth: 01/01/1965
Sex: Female
Medical Record #: MRN-8847612
Accession Number: S24-08934
Specimen Collected: December 10, 2024
Report Date: December 13, 2024
Attending Physician: Dr. Elizabeth Harmon, MD
Department: Breast Surgical Oncology

CLINICAL INFORMATION

55-year-old female with abnormal screening mammogram. Suspicious mass right breast, 10 o'clock position, 5 cm from the nipple. No palpable abnormality. For stereotactic-guided core needle biopsy.

SPECIMEN

A. Core needle biopsy, right breast, 10 o'clock, stereotactic-guided (4 cores)

GROSS DESCRIPTION

Received fresh in formalin labeled with the patient's name, medical record number, and "right breast 10 o'clock" are four tan-pink core needle biopsy tissue fragments ranging from 1.2 to 1.5 cm in length and 0.1 cm in diameter. The specimens are entirely submitted in cassette A1.

MICROSCOPIC DIAGNOSIS

A. Core needle biopsy, right breast, 10 o'clock position:

- **INVASIVE DUCTAL CARCINOMA, no special type**
- **Nottingham histologic grade: 2 of 3**
 - Tubule formation: 2 (10-75% tubules)
 - Nuclear pleomorphism: 2 (moderate variation)
 - Mitotic count: 2 (8 mitoses per 10 HPF; 2.4 mm² area)

- **Associated ductal carcinoma in situ (DCIS):** Present, intermediate nuclear grade, cribriform and solid patterns
- **Lymphovascular invasion:** Not identified in this core biopsy sample

ANCILLARY STUDIES / BIOMARKER PROFILE

Immunohistochemical studies performed on specimen A1:

Marker	Result	Interpretation
Estrogen Receptor (ER)	Positive	Strong nuclear staining in approximately 95% of tumor cells (Allred score 8)
Progesterone Receptor (PR)	Positive	Moderate to strong nuclear staining in approximately 80% of tumor cells (Allred score 7)
HER2 (IHC)	2+ (Equivocal)	Weak to moderate incomplete membrane staining in >10% of tumor cells
HER2 (FISH)	Negative (Not Amplified)	HER2/CEP17 ratio: 1.4 Average HER2 copy number: 3.8 signals/cell (ASCO/CAP guidelines: Non-amplified)
Ki-67 Proliferation Index	22%	Approximately 22% of tumor cell nuclei show positive staining

Summary Classification: Hormone receptor-positive (ER+/PR+), HER2-negative invasive ductal carcinoma.

COMMENT

The core needle biopsy of the right breast demonstrates invasive ductal carcinoma, no special type, Nottingham grade 2 of 3. The tumor is strongly positive for estrogen receptor and progesterone receptor, indicating hormone receptor-positive disease. HER2 testing by immunohistochemistry was equivocal (2+), prompting reflex fluorescence in situ hybridization (FISH) analysis, which confirmed absence of HER2 gene amplification (HER2-negative). The Ki-67 proliferation index of approximately 22% is in the intermediate range.

Clinical correlation is recommended. Definitive surgical excision with assessment of tumor size, margins, and lymph node status is advised for complete staging and treatment planning.

Electronically signed by:

Dr. Robert F. Castellano, M.D.

Board Certified Anatomic and Clinical Pathology

Subspecialty: Breast Pathology

Date/Time: December 13, 2024 14:35 EST

Note: This is a synthetic document created for demonstration purposes. All patient information, clinical data, provider names, and institutional details are fictional. Results and interpretations follow standard pathology reporting guidelines for educational and application testing purposes only.